

1.5.7 Negating Nested Quantifiers

Statements involving nested quantifiers can be negated by successively applying the rules for negating statements involving a single quantifier.

Example 14: Express the negation of the statement $\forall x \exists y (xy=1)$ so that no negation precedes a quantifier.

Soln:

$$\begin{aligned}\neg \forall x \exists y (xy=1) &\equiv \exists x \neg \forall y (xy=1) \quad (\text{De Morgan's law of quantifiers}) \\ &\equiv \exists x \forall y \neg (xy=1) \quad (\text{De Morgan's law of quantifiers}) \\ &\equiv \exists x \forall y (xy \neq 1)\end{aligned}$$

Example 16: Use quantifiers and predicates to express the fact that $\lim_{x \rightarrow a} f(x)$ does not exist where $f(x)$ is a real-valued function of a real variable x and a belongs to the domain of f .

Soln: To say that $\lim_{x \rightarrow a} f(x)$ does not exist means that for all real numbers L , $\lim_{x \rightarrow a} f(x) \neq L$. By using example 8, the statement $\lim_{x \rightarrow a} f(x) \neq L$ can be expressed as $\neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x-a| < \delta \rightarrow |f(x)-L| < \epsilon)$.

Successively applying the rules for negating quantified expressions, we construct this sequence of equivalent statements:

$$\begin{aligned}&\neg \forall \epsilon > 0 \exists \delta > 0 \forall x (0 < |x-a| < \delta \rightarrow |f(x)-L| < \epsilon) \\ &\equiv \exists \epsilon > 0 \neg \exists \delta > 0 \forall x (0 < |x-a| < \delta \rightarrow |f(x)-L| < \epsilon) \quad (\text{De Morgan's law of quantifiers}) \\ &\equiv \exists \epsilon > 0 \forall \delta > 0 \neg \forall x (0 < |x-a| < \delta \rightarrow |f(x)-L| < \epsilon) \quad (\text{De Morgan's law of quantifiers}) \\ &\equiv \exists \epsilon > 0 \forall \delta > 0 \exists x \neg (0 < |x-a| < \delta \rightarrow |f(x)-L| < \epsilon) \quad (\text{De Morgan's law of quantifiers}) \\ &\equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (\neg (0 < |x-a| < \delta) \vee (|f(x)-L| < \epsilon)) \quad (\text{conditional-disjunction equivalence}) \\ &\equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (\neg (0 < |x-a| < \delta) \wedge \neg (|f(x)-L| < \epsilon)) \quad (\text{2nd De Morgan's Law}) \\ &\equiv \exists \epsilon > 0 \forall \delta > 0 \exists x (0 < |x-a| < \delta \wedge \neg (|f(x)-L| < \epsilon)) \quad (\text{By Double negation Law})\end{aligned}$$